Negative Expected Drift Does Not Imply a Unique Attractor at 1: Eliahou Windows, Rhin Linear Forms in Logarithms, and the Collatz Teaming Lemma in Lean 4

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Repository: BenFrohman/collatz-teaming-lemma | Licensed under Apache-2.0

Abstract

We formalize in Lean 4 (`CollatzTeaming`, 23 modules) the inverse-residue "teaming" architecture for the classical Collatz map T, proving the logical equivalence `RemainingLemma` $\leftrightarrow$ `CollatzConjecture`, certifying the trivial cycle $C_0 = (1 \to 4 \to 2 \to 1)$, inverse parent rules, finite orbit scene certificates, and the M_{92} block-map Diophantine identities, while leaving the universal `remaining_lemma` explicitly open. In companion notes (`MU_F_SLOGAN_NOTE.md`, `ELIAHOU_WINDOWS.md`, `RHIN_DERIVATION.md`), we establish the Frohman Model Mean observation: the Syracuse logarithmic drift $\mu_F = \log(3/4) < 0$ is identical for $3n+1$ and $3n-1$, yet $3n-1$ possesses three distinct positive cycles (1, 5, and 17). Thus negative expected logarithmic drift alone cannot rule out non-trivial cycles.

1 Certified Kernel Components vs. Open Frontier

1. **Certified in Lean 4:** The classical map T, trivial cycle $C_0 = \{1, 4, 2\}$, dichotomy/reduction predicates D and R, inverse-residue team definitions, `RemainingLemma` $\leftrightarrow$ `CollatzConjecture`, M_{92} block-map identities, and Eliahou/Steiner/Rhin structural reductions.

2. **Frohman Model Mean Observation ($\mu_F = \log(3/4) < 0$):** Demonstrates via the conjugate map $C_-(n) = 3n - 1$ (equivalent to $3n + 1$ on negative integers) that negative heuristic drift coexists with multiple periodic attractors ($1 \to 2 \to 1$, the 5-cycle at 5, and the 18-step cycle at 17).

3. **Explicit Non-Claim:** No proof of the global Collatz conjecture is claimed; `remaining_lemma` is explicitly marked open.

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A Verification certificate

The formal development was machine-checked as follows.

- Repository:
  `https://github.com/BenFrohman/collatz-teaming-lemma`
- Verified commit:
  `a5152a93fe95743193d403cef4834048f024cfa8`
- Lean toolchain: `leanprover/lean4:v4.34.0`
- Mathlib: none (the development depends only on Lean core)
- Continuous integration run, workflow `certify.yml` (build, axiom audit, `sorry`/`axiom` scan; passed):
  `https://github.com/BenFrohman/collatz-teaming-lemma/actions/runs/41728145753`
- Continuous integration run, workflow `lean.yml` (`lake build`; passed):
  `https://github.com/BenFrohman/collatz-teaming-lemma/actions/runs/41728145773`
- SHA-256 of the GitHub source tarball of the verified commit, downloaded twice with identical result:
  `https://github.com/BenFrohman/collatz-teaming-lemma/archive/`
  `a5152a93fe95743193d403cef4834048f024cfa8.tar.gz`
  `a5dd3007850262e5be31710d1afb0a3bddcd35e3e5c7c6ad8c3a6558f091c4b2`

The tarball hash refers to the verified commit rather than to a release tag, since a document cannot contain the hash of an archive that includes itself. The Lean sources are unchanged between that commit and the release tag.

The development contains no `sorry` and no user-declared axioms. Every one of the 107 theorems and lemmas listed below depends only on Lean's standard axioms `propext`, `Classical.choice` and `Quot.sound` (or on none). The table is the verbatim output of `lake env lean AxiomAudit.lean` (`#print axioms`) on the verified commit.

Lean declaration	`#print axioms`
`CollatzTeaming.remaining_lemma_from_assumption`	`(no axioms)`
`CollatzTeaming.T_even`	`[propext]`
`CollatzTeaming.T_four`	`(no axioms)`
`CollatzTeaming.T_odd`	`[propext]`
`CollatzTeaming.T_one`	`(no axioms)`
`CollatzTeaming.T_two`	`(no axioms)`
`CollatzTeaming.even_double`	`[propext]`
`CollatzTeaming.iter_succ`	`(no axioms)`
`CollatzTeaming.iter_succ'`	`(no axioms)`
`CollatzTeaming.iter_zero`	`(no axioms)`
`CollatzTeaming.iterate_halve`	`[propext]`
`CollatzTeaming.team_maps_to_odd`	`[propext]`
`CollatzTeaming.trivial_cycle`	`(no axioms)`
`CollatzTeaming.two_mul_even`	`[propext]`
`CollatzTeaming.block_n3_k2`	`(no axioms)`
`CollatzTeaming.frohman_block_n3_k2`	`(no axioms)`
`CollatzTeaming.frohman_odd_mod4`	`[propext, Quot.sound]`
`CollatzTeaming.frohman_trivial_block`	`(no axioms)`

Lean declaration	#print axioms
CollatzTeaming.odd_mod4	[propext, Quot.sound]
CollatzTeaming.trivial_block	(no axioms)
CollatzTeaming.forward_even_branch	[propext]
CollatzTeaming.forward_odd_pred	[propext, Quot.sound]
CollatzTeaming.forward_one_double	[propext]
CollatzTeaming.one_reaches_one	(no axioms)
CollatzTeaming.reaches_one_iff_in_reverse_tree	(no axioms)
CollatzTeaming.remaining_forward_iff_reverse	(no axioms)
CollatzTeaming.remaining_lemma_forward	(no axioms)
CollatzTeaming.inclusion_not_binomial_tail	[propext, Classical.choice, Quot.sound]
CollatzTeaming.terras_inclusion	[propext, Classical.choice, Quot.sound]
CollatzTeaming.amber_edge	[propext, Quot.sound]
CollatzTeaming.doubling_parent	[propext]
CollatzTeaming.first_odd_parent	[propext, Quot.sound]
CollatzTeaming.odd_inverse_parent	[propext]
CollatzTeaming.odd_parent_sixteen	[propext, Quot.sound]
CollatzTeaming.orbit27_kernel	(no axioms)
CollatzTeaming.orbit27_length	(no axioms)
CollatzTeaming.orbit27_peak_index	[propext]
CollatzTeaming.orbit_15	(no axioms)
CollatzTeaming.orbit_27	(no axioms)
CollatzTeaming.orbit_7	(no axioms)
CollatzTeaming.orbit_871	(no axioms)
CollatzTeaming.orbit_97	(no axioms)
CollatzTeaming.parentsOf_spec	[propext, Quot.sound]
CollatzTeaming.syracuse_five	(no axioms)
CollatzTeaming.syracuse_not_injective	(no axioms)
CollatzTeaming.syracuse_one	(no axioms)
CollatzTeaming.two_pow_71_reaches_one	[propext]
CollatzTeaming.two_pow_71_value	(no axioms)
CollatzTeaming.C0_closes_one	(no axioms)
CollatzTeaming.reaches_one_of_trivial	(no axioms)
CollatzTeaming.remaining_lemma	(no axioms)
CollatzTeaming.remaining_lemma_iff_collatz	(no axioms)
CollatzTeaming.covering_claim_def	(no axioms)
CollatzTeaming.covering_from_assumption	(no axioms)
CollatzTeaming.C_double	[propext]
CollatzTeaming.C_odd_pred	[propext, Quot.sound]
CollatzTeaming.div_add_mod_six	[propext]
CollatzTeaming.even_preimage	[propext]
CollatzTeaming.four_le_of_mod_six	[propext, Quot.sound]
CollatzTeaming.mod_six_of_odd_step	[propext, Quot.sound]
CollatzTeaming.oddPred_eq	[propext, Quot.sound]
CollatzTeaming.oddPred_odd	[propext, Quot.sound]
CollatzTeaming.oddPredecessor_of_mod_six	[propext, Quot.sound]
CollatzTeaming.one_in_reverse_tree	(no axioms)
CollatzTeaming.remaining_lemma_reverse_tree	(no axioms)
CollatzTeaming.nat_mul_eq_one	[propext]
CollatzTeaming.steiner_L2	(no axioms)
CollatzTeaming.steiner_L_eq_two	[propext, Quot.sound]
CollatzTeaming.steiner_factors	[propext]
CollatzTeaming.steiner_one_cycle	[propext, Quot.sound]
CollatzTeaming.tail_le	[propext, Quot.sound]
CollatzTeaming.tail_zero	(no axioms)
CollatzTeaming.two_pow_le_three_pow	(no axioms)
CollatzTeaming.all_words	[propext]

Lean declaration	#print axioms
CollatzTeaming.badCount_zero_left	(no axioms)
CollatzTeaming.badCount_zero_right	(no axioms)
CollatzTeaming.block_tail	[propext, Quot.sound]
CollatzTeaming.gen_bound	[propext, Quot.sound]
CollatzTeaming.tail_ratio	[propext, Quot.sound]
CollatzTeaming.three_five_lt_two_eight	(no axioms)
CollatzTeaming.two_le_three_pow	[propext, Quot.sound]
CollatzTeaming.bern_243_256	[propext, Quot.sound]
CollatzTeaming.cancel_bound	[propext, Quot.sound]
CollatzTeaming.drop_of_affine	[propext, Classical.choice, Quot.sound]
CollatzTeaming.neverDrops_not_in_tree	[propext, Quot.sound]
CollatzTeaming.not_neverDrops_iff	[propext, Classical.choice, Quot.sound]
CollatzTeaming.ratio_unbounded	[propext, Quot.sound]
CollatzTeaming.residue_count_for_terras	[propext, Quot.sound]
CollatzTeaming.stopping_weaker_than_tree	[propext, Quot.sound]
CollatzTeaming.Block.kpos	(no axioms)
CollatzTeaming.Block.odd	(no axioms)
CollatzTeaming.blockCount_eq	(no axioms)
CollatzTeaming.descent_27	(no axioms)
CollatzTeaming.descent_703	(no axioms)
CollatzTeaming.four_reaches_trivial_at_zero	(no axioms)
CollatzTeaming.missing_term_type	(no axioms)
CollatzTeaming.one_reaches_trivial_at_zero	(no axioms)
CollatzTeaming.one_zero_path	(no axioms)
CollatzTeaming.reaches_trivial_at_zero	(no axioms)
CollatzTeaming.two_reaches_trivial_at_zero	(no axioms)
CollatzTeaming.witness_shape_of_remaining	(no axioms)
CollatzTeaming.zero_fixed	(no axioms)
CollatzTeaming.zero_orbit	(no axioms)
CollatzTeaming.zero_path	(no axioms)
CollatzTeaming.zero_path_four	(no axioms)
CollatzTeaming.zero_path_one	(no axioms)
CollatzTeaming.zero_path_two	(no axioms)

Separately built modules

Four modules redeclare names already present in the root library and so cannot be imported alongside it: `ParentRule`, `Scenes`, `TerrasDensity` and `Derived`. Each is built on its own with `lake build CollatzTeaming.`*Module* and audited by its own file `AxiomAudit_`*Module*`.lean` in the same CI run. All 41 declarations (4, 17, 1 and 19 respectively) are free of `sorry` and depend only on the standard axioms. In total 148 declarations are audited. Orbit-table proofs in `Scenes` and `Derived` use `decide +kernel` (kernel evaluation, no extra axioms); `native_decide` is not used anywhere.

`ParentRule`.

Lean declaration	#print axioms
CollatzTeaming.doubling_parent	[propext]
CollatzTeaming.iterate_halve_parent	[propext]
CollatzTeaming.odd_inverse_parent	[propext]
CollatzTeaming.parentsOf_spec	[propext, Quot.sound]

`Scenes`.

Lean declaration	#print axioms
CollatzTeaming.amber_edge	(no axioms)
CollatzTeaming.five_is_odd_parent_of_sixteen	(no axioms)
CollatzTeaming.head_7	(no axioms)
CollatzTeaming.odd_parent_sixteen	[propext, Quot.sound]
CollatzTeaming.orbit_15	(no axioms)
CollatzTeaming.orbit_255	(no axioms)
CollatzTeaming.orbit_27	(no axioms)
CollatzTeaming.orbit_6171	(no axioms)
CollatzTeaming.orbit_7	(no axioms)
CollatzTeaming.orbit_703	(no axioms)
CollatzTeaming.orbit_871	(no axioms)
CollatzTeaming.orbit_97	(no axioms)
CollatzTeaming.sixteen_mod_six	(no axioms)
CollatzTeaming.syracuse_five	(no axioms)
CollatzTeaming.syracuse_not_injective	(no axioms)
CollatzTeaming.syracuse_one	(no axioms)
CollatzTeaming.two_pow_71_reaches_one	[propext]

TerrasDensity.

Lean declaration	#print axioms
CollatzTeaming.terrasDensity_statement	(no axioms)

Derived.

Lean declaration	#print axioms
CollatzTeaming.odd_inverse_parent	[propext]
CollatzTeaming.first_odd_parent	(no axioms)
CollatzTeaming.first_odd_parent_rule	[propext]
CollatzTeaming.even_parent	[propext]
CollatzTeaming.syracuse_one	(no axioms)
CollatzTeaming.syracuse_five	(no axioms)
CollatzTeaming.syracuse_not_injective	(no axioms)
CollatzTeaming.orbit_7_peak	(no axioms)
CollatzTeaming.orbit_7_reaches	(no axioms)
CollatzTeaming.orbit_27_peak	(no axioms)
CollatzTeaming.orbit_27_reaches	(no axioms)
CollatzTeaming.orbit_97_peak	(no axioms)
CollatzTeaming.orbit_97_reaches	(no axioms)
CollatzTeaming.orbit_871_peak	(no axioms)
CollatzTeaming.orbit_871_reaches	(no axioms)
CollatzTeaming.orbit_6171_peak	(no axioms)
CollatzTeaming.orbit_6171_reaches	(no axioms)
CollatzTeaming.orbit_two_pow_71	[propext]
CollatzTeaming.two_pow_71_value	(no axioms)

A removed statement

An earlier version of `Derived` contained the theorem

$$\texttt{derived_does_not_close_collatz} : (\texttt{iter 111 27 = 1}) \neq \texttt{CollatzConjecture},$$

intended to record that finitely many orbit checks do not prove the Collatz conjecture. It does not formalize that claim. Since `iter 111 27 = 1` is proved (`orbit_27_reaches`), it equals `True` as a proposition by propositional extensionality, so the statement is equivalent to $\neg$ `CollatzConjecture`:

unprovable unless the conjecture is false, and false if it is true. The intended assertion is about provability and is not expressible as a proposition of this form. The theorem has therefore been replaced by an explanatory comment; nothing in the development proves or refutes the conjecture.